

Sanjay Acharjee

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Physical AI | Field Robotics | 3D Perception | Active Perception | Mobile Manipulation | Spatial Intelligence

Summary

Ph.D. researcher developing robotic perception, spatial intelligence, and autonomy for complex, changing, and unstructured environments. Research spans active perception, mobile manipulation, metric 3D reconstruction, Gaussian splatting, visual servoing, reality capture, computer vision, and multimodal AI. Hands-on work includes **Husky A300 + UR5e**, **LeRobot SO-101**, ROS 2-based autonomy, Gazebo/MuJoCo simulation, RGB-D/LiDAR sensing, and deployment of perception/control pipelines to physical robots.

Education

The University of Texas at Arlington

Ph.D. in Civil Engineering, CGPA: 4.00/4.00

Arlington, TX

Jan. 2024 – Expected Fall 2027

- **Graduate Certificate:** Artificial Intelligence & Deep Learning, May 2026.
- **Relevant Coursework:** Robotics, Neural Networks, Computer Vision, Machine Learning, Artificial Intelligence, Data Analysis & Modeling, and Advanced Building Information Modeling.

Bangladesh University of Engineering and Technology (BUET)

B.Sc. in Civil Engineering, CGPA: 3.44/4.00

Dhaka, Bangladesh

May 2023

Technical Skills

Programming & Systems: Python, C++, CUDA/C++, MATLAB, Git, Linux/Ubuntu

Robotics: ROS 2, Nav2, TF2, URDF/Xacro, MoveIt, forward/inverse/differential kinematics, Jacobian control, visual servoing, path planning, A*/Dijkstra

Simulation & Robot Learning: Gazebo, MuJoCo, Hugging Face LeRobot, Sim2Real workflows

AI / Computer Vision: PyTorch, TensorFlow, OpenCV, Hugging Face, transformers, vision-language models, diffusion models, pose estimation, text-grounded segmentation

3D Perception & Spatial Computing: 3D Gaussian Splatting, COLMAP, Open3D, point clouds, SfM/MVS, RGB-D sensing, LiDAR, camera geometry, metric 3D reconstruction, mesh processing

Digital Construction & Reality Capture: BIM/VDC, Scan-to-BIM, Autodesk Revit, Dynamo, IfcOpenShell, Trimble RealWorks

Hardware: Clearpath Husky A300, UR5e, LeRobot SO-101, RGB/RGB-D cameras, Trimble X9, Parrot Anafi USA, Autel Evo Max 4T

Research Experience

University of Texas at Arlington – iSET Lab

Graduate Research Assistant

Arlington, TX

Jan. 2024 – Present

- **DeltaSeek – Active Perception / Mobile Manipulation:** Developed an embodiment-aware sensing framework using a **Husky A300 + UR5e**. Evaluated **240 navigable base poses** (60 positions $\times$ 4 yaw orientations) across five articulated arm postures; wrist-mounted sensing observed changes inaccessible from chassis viewpoints and reduced median travel from **15.2 m to 6.0 m** for mutually observable changes. **Paper | Code**
- **TwinSplat – Metric 3D Gaussian Splatting / Spatial Intelligence:** Developing a Gaussian-based indoor digital-twin system that combines photorealistic novel-view rendering with dense metric surface recovery from a single scene representation. Across three real indoor scenes evaluated against survey-grade terrestrial LiDAR, achieved up to **0.759 Geometry F1@5 cm**, **1.7 cm median surface error**, and **191 FPS** full-resolution rendering. **Public Showcase**
- **TwinSplat Application Layer:** Built language-guided 3D localization that grounds free-text object queries across multiple rendered viewpoints into persistent 3D object instances and navigation targets; implemented a browser-based visualization stack using Flask/three.js, mesh raycasting, and cross-view instance association. Research code, ablations, and full technical methodology remain private while the work is developed toward conference submission.
- **Physical AI / Field Robotics Platform:** Developing a camera-equipped **Husky A300 + UR5e** platform using ROS 2, Nav2, TF2, Gazebo, MoveIt, navigation costmaps, and robot/sensor integration for closed-loop active perception, autonomous navigation, 3D sensing, and mobile manipulation in changing environments.
- **3D Multimodal Perception:** Developed an SfM/MVS pipeline fusing RGB and thermal drone imagery to reconstruct dense 3D thermal profiles for spatially localized building inspection and anomaly detection using COLMAP and Open3D.
- **Generative / Multimodal AI:** Developed scene-graph-guided pipelines for extracting industrial hazards from historical accident narratives and generating/evaluating visual safety scenarios using LLMs, vision-language models, and diffusion models.

AIEdgeInside

Machine Learning Engineer – Full-time

Remote, Tokyo, Japan

Jun. 2023 – Dec. 2023

- Developed computer-vision/OCR pipelines for handwritten table documents using transformer-based models including TrOCR, automating cell-level information extraction.
- Developed generative-image pipelines for virtual try-on applications and realistic apparel synthesis.

Selected Robotics & 3D Projects

LeRobot SO-101 Kinematics & Visual Servoing | *MuJoCo, LeRobot, OpenCV, NumPy, SymPy*

2025–2026

- Built a robotics control stack with analytical forward kinematics, Jacobian-based differential inverse kinematics, singularity monitoring, predictive joint-limit safety, and closed-loop image-based visual servoing.
- Validated algorithms in **MuJoCo** before deploying the same control framework to the physical **LeRobot SO-101**; implemented perception-to-control pipelines that convert image-space target errors into joint commands for autonomous end-effector positioning.

GitHub

TwinSplat – Indoor Metric Digital Twins from Gaussian Splatting | *PyTorch, CUDA/C++, COLMAP, Open3D, OpenCV*, 2026
three.js

- Evaluated rendering with PSNR/SSIM/LPIPS and geometry against survey-grade LiDAR using accuracy, completeness, precision, recall, F1, and surface-error metrics; standard 3DGS geometry F1 improved from 0.192 to **0.759** (classroom), 0.473 to **0.616** (lobby), and 0.076 to **0.716** (lounge).
- Brought a research CUDA rasterizer onto a modern CUDA 12.x toolchain, removed a heavyweight dependency from the training path, extended data loading for the indoor dataset, and instrumented rendering for frame-rate and GPU-memory benchmarking.

3D Thermal Profiling from Multimodal Drone Data | *COLMAP, Open3D, SfM/MVS, RGB + LWIR*

2024–2025

- Reconstructed building-scale 3D geometry from drone imagery and mapped thermal information into dense point clouds for spatially localized inspection and energy diagnostics.

Teaching, Reality Capture & UAS

University of Texas at Arlington

Arlington, TX

Graduate Teaching Assistant – Advanced Building Information Modeling

2024 – 2025

- Delivered technical instruction in **Dynamo-based BIM automation** and **Reality Capture / Scan-to-BIM**; led hands-on **Trimble X9 LiDAR** workflows covering point-cloud acquisition, Trimble RealWorks processing, and integration into Autodesk Revit.
- **Additional GTA role – CE 4334: Drones & Advanced Construction Technology**: Trained students in safe drone operation and construction field-data collection, including **FAA Part 107 compliance**, pre-flight planning, airspace/safety considerations, and supervised flight exercises.

Selected Publications & Presentations

- **Acharjee, S., Sakib, M. N. DeltaSeek: Toward Active Perception in Evolving Construction Environments.** arXiv:2609.33836; IROS Construction Robotics Workshop, 2026. **Paper** | **Code**
- **Acharjee, S., Khan Ratul, A., Patiño, D., Sakib, M. N. Floorplan Reconstruction through Field Prediction and Geometry-Aware Vectorization.** Submitted to AAAI 2027.
- **Khan Ratul, A., Acharjee, S., Park, S., Sakib, M. N. Multi-agent human-AI collaborative framework for converting hand-drawn floor plans to 3D BIM.** *Automation in Construction*, 2026.
- **Acharjee, S., Khan Ratul, A., Patiño, D., Sakib, M. N. Scene Graph-Guided Generative AI Framework for Synthesizing and Evaluating Industrial Hazard Scenarios.** arXiv:2511.13970, 2025.
- **Acharjee, S., Khan Ratul, A., Aryal, A., Sakib, M. N. SfM Meets Thermal: Reconstructing 3D Thermal Profiles of Buildings Using Multimodal Drone Data.** ASCE International Conference on Computing in Civil Engineering, 2025.
- **Acharjee, S., Khan Ratul, A., Aryal, A., Sakib, M. N. Reconstructing 3D Thermal Profiles of Buildings Using Multimodal Drone Data.** Poster presented at the *4th Workshop on Future of Construction: Safe, Reliable, and Precise Robots in Construction Environments*, IEEE International Conference on Robotics and Automation (ICRA), 2025. **Poster**
- **Acharjee, S., Khan Ratul, A., Patiño, D., Sakib, M. N. Data-Driven Safety: Leveraging Historical Accident Data and Generative AI for Generating Workplace Hazard Visualizations.** ASCE International Conference on Computing in Civil Engineering, 2025.
- **Nahin, S. K., Acharjee, S., Saha, S., Das, A., Hossain, S., Haque, M. A. Human sleeping pose estimation from IR images for in-bed patient monitoring using image processing and deep learning techniques.** *Heliyon*, 2024.

Honors & Certifications

- **Champion**, IEEE Video and Image Processing Cup (VIP Cup), 2021.
- **Second Runner-up**, IEEE Video and Image Processing Cup, 2022.
- **First Runner-up**, DL Sprint Bengali Automatic Speech Recognition Competition, 2022.
- **Graduate Dean's STEM Research Assistantship**, University of Texas at Arlington, 2024 and 2026.
- **FAA Part 107 Remote Pilot Certificate (sUAS).**